

Co-simulation: State of the art

C Gomes, C Thule, D Broman, PG Larsen... - arXiv preprint arXiv ..., 2017 - arxiv.org

... We provide an **overview** of **co-simulation** approaches, research challenges, and research op... with a detailed **taxonomy** with different aspects of the **state** of the **art** of **co-simulation** and ...

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Co-simulation: a survey

C Gomes, C Thule, D Broman, PG Larsen... - ACM Computing ..., 2018 - dl.acm.org

... nature, **cosimulation** is being studied in different disciplines but with limited sharing of findings. In this **survey**, we study and **survey** the **state-of-the-art** techniques for **co-simulation**, with ...

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An empirical survey on co-simulation: Promising standards, challenges and research needs

G Schweiger, C Gomes, G Engel, I Hafner... - ... modelling practice and ..., 2019 - Elsevier

... Then, in Step (2), the academia category was populated based on a keyword-based search in the literature on the **state** of the **art** in **co-simulation** (see Section 1.1). The industry ...

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Co-simulation-based optimization: state of the art

DA Vega, V Chevrier - 2024 - inria.hal.science

... The emerging behavior achieved with the **co-simulation** can ... Introducing optimization as the next step of **co-simulation** ... we present a **survey** on the combination of **co-simulation** and ...

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Co-Simulation-An Empirical Survey: Applications, Recent Developments and Future Challenges.

G Schweiger, GP Engel, JP Schöggel, I Hafner... - Simul. Notes ..., 2020 - sne-journal.org

... An **overview** of **co-simulation** approaches and ... **survey** aims to merge different views of heterogeneous communities which are working in the field of **co-simulation**, on the **state** of the **art**, ...

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Co-simulation for buildings and smart energy systems—A taxonomic review

Q Alfalouji, T Schranz, B Falay, S Wilfling... - ... Modelling Practice and ..., 2023 - Elsevier

... By analyzing the results of a taxonomic **review** and comparing the results with other **state-of-the-art** reviews, we identify research gaps and propose best practices for **co-simulation** ...

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A survey and statistical analysis of smart grid co-simulations

M Vogt, F Marten, M Braun - Applied energy, 2018 - Elsevier

... Definition of **state-of-the-art co-simulation** and provision of recent trends. ... **survey** of 26 smart grid **co-simulation** frameworks. First of all, we present our understanding of a **co-simulation**. ...

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Co-simulation: the past, future, and open challenges

C Gomes, C Thule, J Deantoni, PG Larsen... - ... Applications of Formal ..., 2018 - Springer

... kinds of formalisms (eg, **state** machine models for software-based ... **Co-simulation** has been proposed as a way forward by ... an historical **overview** of the different facets of **co-simulation**; ...

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Smart grids co-simulations: **Survey & research directions**

P Mihal, M Schvarcbacher, B Rossi, T Pitner - ... Computing: Informatics and ..., 2022 - Elsevier

... To classify **co-simulation** articles in this **review**, we used a classification of SG research in nine main areas (A1–A9) as defined in Cintuglu et al. [7], in which the last two areas (A8. ...

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Co-simulation platforms for co-design of networked control systems: **An overview**

W Li, X Zhang, H Li - Control Engineering Practice, 2014 - Elsevier

... given to show the beneficial of **co-simulation** approaches. ... Section 3 introduces and compares the **state-of-the-art** ... a brief **survey** of the existing integrated **co-simulation** platforms...

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Assessing the need and **state** of power system and transportation system **co-simulation**

JMB Smith, K Balasubramaniam, R Hadidi - IEEE Access, 2023 - [ieeexplore.ieee.org](#)

... require **cosimulation** to integrate them with the electrical simulators. Gomes provides a **state** of the **art** of **co-simulation** in [13] which holistically showcases the broad **state** without a ...

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Future perspectives of **co-simulation** in the smart grid domain

C Steinbrink, F Schlögl, D Babazadeh... - 2018 IEEE ..., 2018 - [ieeexplore.ieee.org](#)

... Even though **co-simulation** is gaining more and more momentum, experience shows that ... **state** of the **art** of **co-simulation** in CPES is concisely presented and analyzed. Related **review** ...

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A comparison study of **co-simulation** frameworks for multi-energy systems: the scalability problem

L Barbierato, P Rando Mazzarino, M Montarolo... - Energy ..., 2022 - Springer

... This figure offers a general **overview** of the interacting components in a **co-simulation** framework, while Fig. 2b–d represent the specific implementations of these components inside the ...

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Ten questions concerning **co-simulation** for performance prediction of advanced building envelopes

E Taveres-Cachat, F Favoino, R Loonen... - Building and ..., 2021 - Elsevier

... these difficulties has been to use **co-simulation**. **Co-simulation** allows modelers to use multiple ... This article provides a critical **overview** of the possibilities that **co-simulation** approaches ...

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On the terminology and structuring of **co-simulation** methods

I Hafner, N Popper - Proceedings of the 8th International Workshop on ..., 2017 - [dl.acm.org](#)

... differences in terminology arising from these various origins, the levels on which they meet as well as an attempt on the classification and structuring of current **state-of-the-art** methods. ...

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Co-simulation de systemes complexes avec le framework CoSim20

G LIBONI - 2021 - [researchgate.net](#)

... We discussed the main challenges addressed in this thesis, the assumptions we took, and the limitations encountered and identified in the current **state** of the **art**. Chapter ...

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Towards a classification scheme for **co-simulation** approaches in energy systems

F Schloegl, S Rohjans, S Lehnhoff... - ... on Smart Electric ..., 2015 - [ieeexplore.ieee.org](#)

... It is then possible to set the fixed step simulator back to the **state** when the event of the event ... physical and functional parts according to the current **art** of science, we receive a white box ...

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Towards generic SCADA simulators: A **survey** of existing multi-purpose **co-simulation** platforms, best practices and use-cases

K Mathioudakis, N Frangiadakis... - Scientific ..., 2013 - [conf-scoop.org](#)

... is to **review** the **state** of the **art** of ... a **co-simulation** platform should be in line. In Section IV we describe and analyze the communication network **co-simulation** tools as well as we **survey** ...

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A model-driven **co-simulation** environment for heterogeneous systems

M Bombino, P Scandurra - International Journal on Software Tools for ..., 2013 - Springer

... This article proposes a model-driven continuous/discrete **co-simulation** framework based on

... The proposed approach adopts a code-in-the-loop **co-simulation** schema where optimized ...

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An Overview of the State of the Art in Co-Simulation and Related Methods.

I Hafner, N Popper - Simul. Notes Eur., 2021 - sne-journal.org

... [141] provide a **survey** on **state-of-the-art** techniques for **co-simulation**, starting by the

introduction of a formalization similar to DEVS ([76]). As challenges specific to DE **co-simulation**, [...

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Multi-domain simulation utilizing SysML: **state** of the **art** and future perspectives

C Nigischer, S Bougain, R Riegler, HP Stanek... - *Procedia CIRP*, 2021 - Elsevier

... [4] provide an **overview** of SysML simulation approaches and the accompanying challenges.

... **co-simulation** and new evolving standards like the Functional Mock-up Interface (FMI). ...

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Moving from **co-simulation** to simulation for effective smart systems design

F Fummi, M Lora, F Stefanni... - ..., *Automation & Test ...*, 2014 - ieeexplore.ieee.org

... allows to identify a precise role in the design flow for **co-simulation** and simulation scenarios.

... Employing **state** of the **art** techniques, such as heterogeneous **co-simulation**, leads to a ...

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Interfacing power system and ICT simulators: Challenges, **state-of-the-art**, and case studies

SC Müller, H Georg, JJ Nutaro, E Widl... - ... on Smart Grid, 2016 - ieeexplore.ieee.org

... Thus, it is impossible to dictate a best solution, and even comparisons of solutions for **co-simulation** are difficult if we do not restrict ourselves to very narrow use cases. Therefore, we ...

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An automated methodology to select functional **co-simulation** configurations

J Rahikainen, F González, MÁ Naya - *Multibody System Dynamics*, 2020 - Springer

... Figure 1 shows the model used in this research, namely two subsystems, 1 and 2, coupled

in a **co-simulation** setup through a **co-simulation** manager. Each subsystem has its own set of ...

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Real-time **co-simulation** for the virtual commissioning of production systems

C Scheifele, A Verl, O Riedel - *Procedia CIRP*, 2019 - Elsevier

... The real-time **co-simulation** presented in this paper promises a much greater significance of digital twins. Thus, real-time **co-simulation** increases the possibilities of the virtual ...

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A compilation-based software estimation scheme for hardware/software **co-simulation**

M Lajolo, M Lazarescu... - *Proceedings of the ...*, 1999 - dl.acm.org

... -task basis by attaching parameters to tasks in our **co-simulation** framework. To the best of our

... that combines a **state-of-the-art** optimizing compiler with a high-level **co-simulation** and co-...

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Hybrid modeling of production systems: **co-simulation** and DEVS-based approach

B Heinzl - 2016 - repositum.tuwien.at

... of a **co-simulation**. ... **state-of-the-art** simulation tools from research literature are

employed: BCVTB (Building Controls Virtual Test Bed) provides a middleware solution for **co-simulation**, ...

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Vico: An entity-component-system based **co-simulation** framework

LI Hatledal, Y Chu, A Styve, H Zhang - *Simulation Modelling Practice and ...*, 2021 - Elsevier

... This paper introduces a novel **co-simulation** framework running ... Mock-up Interface standard

for **co-simulation**, as well as the ... A recent **survey** showed that experts consider the FMI ...

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Hybrid simulation modelling in operational research: A **state-of-the-art** review

SC Brailsford, T Eldabi, M Kunc, N Mustafee... - European Journal of ..., 2019 - Elsevier

... Given the importance of this emerging area and its relevance to operational research, this paper provides a **review** of the topic from an OR perspective. The results of a **review** of the ...

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Towards a **co-simulation** based model assessment process for system architecture

B Bossa, B Boulbene, S Dubé, M Pantel - ... Technologies: Applications and ..., 2018 - Springer

... However, the current **state** of the **art** is not sufficient to achieve a seamless use in an Extended Enterprise (EE) context. Indeed, the various stakeholders must protect their Intellectual ...

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Maestro: The INTO-CPS **co-simulation** framework

C Thule, K Lausdahl, C Gomes, G Meisl... - ... Modelling Practice and ..., 2019 - Elsevier

... All the other FMU participating in the **co-simulation** are then rolled back to the previous **state** and prompted to perform a step of size H' . If a fixed step size has been set, Maestro will ...

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Complex systems **co-simulation** with the CoSim20 framework: for efficient and accurate distributed co-simulations

G Liboni - 2021 - theses.hal.science

... We discussed the main challenges addressed in this thesis, the assumptions we took, and the limitations encountered and identified in the current **state** of the **art**. Chapter ...

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Model-based configuration of automotive **co-simulation** scenarios

M Krammer, J Fritz, M Karner - Proceedings of the 48th Annual ..., 2015 - dl.acm.org

... **Co-simulation** represents a **state-of-the-art** method for virtual ... setup of all involved **co-simulation** tools within a framework ... the idea of a model based **co-simulation** configuration further, ...

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Features of integrated model-based co-modelling and **co-simulation** technology

PG Larsen, J Fitzgerald, J Woodcock, C Gamble... - ... Engineering and Formal ..., 2018 - Springer

... analysis by means of **co-simulation** based on a reconciled ... The idea is to enable **co-simulation** of extensible groups of ... In this paper, we first provide an **overview** of what we consider ...

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PyCARL: A PyNN interface for hardware-software **co-simulation** of spiking neural network

A Balaji, P Adiraju, HJ Kashyap, A Das... - arXiv preprint arXiv ..., 2020 - arxiv.org

... programming interface for hardware-software **cosimulation** of spiking neural network (SNN). ...

Second, we integrate cycle-accurate models of **state-of-the-art** neuromorphic hardware ...

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Co-simulation for performance prediction of innovative integrated mechanical energy systems in buildings

M Trcka - 2008 - research.tue.nl

... generic **co-simulation** implementation are defined. The main results of the research show that **co-simulation** facilitates rapid expansion of modeling capabilities of the **state** of the **art** BPS ...

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Facilitating Smart Grids Integration Through a Hybrid Multi-Model **Co-Simulation** Framework

L Barbierato, DS Schiera, M Orlando, A Lanzini... - IEEE ..., 2024 - ieeexplore.ieee.org

... To summarize, the proposed platform pushes the boundaries of the current **state** of the **art** by integrating literature and novel technologies to develop a design and simulation framework ...

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Co-simulation of building energy and control systems with the Building Controls Virtual Test Bed

M Wetter - Journal of Building Performance Simulation, 2011 - Taylor & Francis

... model used to implement the **co-simulation** is discussed. The ... in which different **state** of the **art** simulation programs are linked ... a solution to the coupled problem using **co-simulation**. ...

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[Towards the verification of hybrid **co-simulation** algorithms](#)

[C Thule, C Gomes, J Deantoni, PG Larsen... - ... and Foundations: STAF ..., 2018 - Springer](#)

... **co-simulation** of a family of hybrid systems can fail to reproduce the order of events that should have occurred (event ordering); (ii) how to prove that a **co-simulation** ... the **state** of the **art**. ...

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[COSCO: Container orchestration using **co-simulation** and gradient based optimization for fog computing environments](#)

[S Tuli, SR Poojara, SN Srirama... - ... on Parallel and ..., 2021 - ieeexplore.ieee.org](#)

... **Co-simulation** and the back-propagation ... time, Service Level Objective and scheduling time by up to 15, 40, 4, and 82 percent respectively when compared to the **state-of-the-art** ...

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Co-simulation training platform for smart grids

T Strasser, M Stifter, F Andr  n... - IEEE Transactions on ..., 2014 - ieeexplore.ieee.org

... paper is the introduction of a **co-simulation**-based platform for ... to cope with complexity by the **co-simulation** of components. A ... follows: Section II provides an **overview** of the related work. ...

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A categorical model of process co-simulation

DC Craciunean, D Karagiannis - International Journal of ..., 2019 - pdfs.semanticscholar.org

... **cosimulation** model based on the categorical sketch and in section 4 we will see how the **co-simulation** ... A **state** $\mathfrak{S}k$ of the **co-simulation** model is a tuple of the form $\mathfrak{S}k=(\ ,\ ,\ ,\ ,\ ,\ ,\)$ where $\in ob...$

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Synchrosim: An integrated co-simulation middleware for heterogeneous multi-robot system

E Dey, J Hossain, N Roy... - 2022 18th International ..., 2022 - ieeexplore.ieee.org

... To measure the communication performance, we have chosen one of the **state-of-the-art** network simulators, NS-3 which is compatible with the selected wireless stack. To integrate both ...

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A framework for co-simulation based optimization

DA Vega, V Chevrier - Proceedings of the 35th European Modeling & ..., 2023 - hal.science

... This section contains a literature **survey** that reveals some general patterns that could guide any **co-simulation** user to include an optimization process in a complex sys- ...

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HintCO: Hint-based Configuration of Co-simulations

C Gomes, B Oakes, M Moradi... - Proceedings of the ..., 2019 - repository.uantwerpen.be

... These hints, together with **state-of-the-art** best practices, are then used to semi-automatically guide the configuration process of the **co-simulation**. We report the application of this ...

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Analysis of coupling strategies and protocols for co-simulation

C P  hringer - 2017 - scholar.archive.org

... Due to the current developments in the area of Cyber-Physical Systems, **state-of-the-art** research in this field is largely based on **co-simulation** between continuous and discrete ...

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ZerOBNL: A framework for distributed and reproducible co-simulation

P Puerto, E Widl, J Page - 2019 7th Workshop on Modeling ..., 2019 - ieeexplore.ieee.org

... Abstract—**Co-simulation** combines multiple tools depending ... environments and makes **co-simulation** difficult to deploy and ... **co-simulation** is presented, based on **state-of-the-art** open-...

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Effective processor verification with logic fuzzer enhanced co-simulation

N Kabylkas, T Thorn, S Srinath, P Xekalakis... - MICRO-54: 54th Annual ..., 2021 - dl.acm.org

... We also present Dromajo, the **state** of the **art** processor verification framework for RISC-V cores. Dromajo is an RV64GC emulator that was designed specifically for **co-simulation** ...

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Game theory-based ramp merging for mixed traffic with unity-sumo **co-simulation**

X Liao, X Zhao, Z Wang, K Han, P Tiwari... - ... on Systems, Man ..., 2021 - ieeexplore.ieee.org

... **state-of-the-art** microscopic traffic simulator. The results show that the average speed of traffic flow can be increased up to 210%, while the fuel consumption can be reduced up to 53.9%...

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Native ISS-SystemC integration for the **co-simulation** of multi-processor SoC

F Fummi, S Martini, G Perbellini... - ... , Automation and Test ..., 2004 - ieeexplore.ieee.org

... is the core of **state-of-the-art co-simulation** approaches. The efficiency of the **co-simulation** is

... In this work, we have proposed two **co-simulation** schemes, that differ in how the integration ...

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